GREEN ROOF THERMAL SIMULATION

Current Working Documentation and Validation Plan

CAM (Bromelia) vs C3 (Wedelia)

Finite Volume Method | Chagolla-Aranda et al. reference framework

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Status statement
This document is prepared as a current progress documentation for professor discussion. The CAM result is not framed as final validation yet. The strongest current result is a path-aware diagnostic: T2A is the most usable temporary damped-path target, while T1Tb and T1Ta require physical sensor/path verification.

Prepared from the current CAM v7 workflow, stack-aware validation output, and C3 quick-validation plan.

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Chapter 1: Overview & Current Position

This documentation summarizes the current state of the green roof thermal simulation comparing CAM (Bromelia) and C3 (Wedelia) systems. The implementation follows a coupled heat and moisture finite-volume framework adapted from the reference green roof model, with local experimental data from weather station, NI temperature sensors, Licor, and soil moisture sensors where available.

Current technical position

The current task is not to claim a perfect final validation. The current task is to show meaningful progress: (1) the model pipeline runs, (2) stack-aware validation has been added, (3) the original T1Tb target is not physically safe to force-fit, and (4) T2A is the most defensible temporary damped-path validation target for CAM until sensor mapping is physically verified.

1.1 Current Research Framing

Item	Current statement
Research object	Green roof thermal response using CAM and C3 vegetation types in tropical-humid Jakarta context.
Model type	1D coupled heat and moisture model using finite volume discretization for canopy, substrate, and slab layers.
Current validation stage	Path-aware diagnostic validation, not final proof of one-to-one physical node mapping.
Most useful CAM target now	T2A vs model T_s_in, interpreted as temporary damped-path validation.
Problematic CAM targets	T1Tb vs T_s_in and T1Ta vs T_s_top show large amplitude/path mismatch.
Immediate experimental TODO	Plug-off sensor test to confirm physical channel mapping.

1.2 Updated Novelty Claims

Claim	Status	Current phrasing
N1: CAM vs C3 comparison	Strong	Direct comparison of CAM and C3 plant thermal behavior using the same local green roof setup and model framework.
N2: Local substrate	Strong	Use of local soil plus rice husk substrate rather than imported lightweight aggregate.
N3: Dynamic indoor boundary	Strong	Indoor/box air temperature is driven by NI data, not constant 25 degC from air-conditioned assumption.
N4: Path-aware validation	New/important	Measured sensors are not assumed to be exposed surfaces; validation is performed against physical stack positions and alternative heat-path candidates.
N5: Sensor/path diagnostic	New/important	Large mismatch in T1Ta and T1Tb is interpreted as evidence for sensor/path verification, not blindly tuned parameters.

Chapter 2: Experimental Data & Sensor Context

2.1 Data Sources

Source	Availability	Variables / Role	Current use
Davis / weather file	Available	T_a, RH, wind speed, rain, G_sol	Boundary forcing for model and solar-response diagnostics.
NI temperature sensors	Available	Multi-channel thermocouple/temperature sensors at 1-minute grid after resampling	Main measured target and boundary data.
Soil moisture sensors	Available for short windows	Sensor 1 and sensor 2, shallow/deep inference	Used for CAM diagnostic and constraint reasoning, not yet final full-period forcing.
Licor 6800 C3	Available but discontinuous	gsw, A, Tleaf, energy balance variables	Used for stomatal resistance estimates and physiology justification.
Licor 6800 CAM	Older and discontinuous	CAM gsw ranges, including low daytime values and similar day/night percentiles in some sessions	Used carefully as physiological evidence, not continuous forcing.

2.2 Updated Physical Stack Interpretation

The key clarification is that there is no sensor assumed to be truly exposed to direct outdoor radiation. Therefore, high solar correlation should be interpreted as a fast/upper heat-path response, not proof that the sensor is directly exposed.

Physical position	Expected model node	Current default measured channel	Status
Tanah atas / upper substrate	T_g_top	T1Ke	Physical mapping plausible, but current model T_g_top is too hot and poorly correlated.
Tanah bawah / lower substrate	T_g_bot	T2A	Physical mapping plausible, but T2A dynamically matches T_s_in better than T_g_bot.
Atap outdoor / top slab under substrate	T_s_top	T1Ta	v7 now exports T_s_top; current match is poor and measured T1Ta is much more dynamic.
Atap indoor / bottom slab / inner roof	T_s_in	T1Tb	Current measured amplitude much larger than model; not safe as main target.
Ruangan / indoor air boundary	T_in_used	T1Ka	Excellent match, because it is used as dynamic boundary.

Practical implication

Sensor names from older nomenclature should not be treated as absolute truth for current March-April data. The physical plug-off test is still required to confirm channel mapping.

Chapter 3: Code Architecture

The current workflow is separated into model simulation, stack-aware validation, quick C3 validation, and professor-ready plot generation. The purpose of the separation is to avoid hiding diagnostics inside calibration logic.

3.1 Current Pipeline Flowchart

```
PHASE 1: Inputs and fixed parameters
  INIT plant parameters: CAM and C3 canopy/leaf/gsw assumptions
  INIT substrate parameters: soil + rice husk properties
  INIT slab parameters: concrete slab, H_slab, h_in
  INIT weather + NI + soil moisture files

PHASE 2: Run CAM v7 model
  DO run_cam_gsw_driven.py
  OUTPUT T_f, T_g_top, T_g_mid, T_g_bot, T_s_top, T_s_mid, T_s_in, T_in_used

PHASE 3: Stack-aware CAM validation
  DO cam_stack_aware_validation.py
  COMPARE physical pairs: T1Ke-T_g_top, T2A-T_g_bot, T1Ta-T_s_top, T1Tb-T_s_in, T1Ka-T_in_used
  OUTPUT stack metrics, pair time series, diagnostic plots

PHASE 4: Professor CAM plot pack
  DO make_cam_professor_plots.py
  OUTPUT T2A vs T_s_in main plot, T1Tb diagnostic plot, summary table

PHASE 5: Quick C3 validation
  DO run_c3_quick_validation.py
  FIND best candidate sensor for requested C3 model node, usually T_s_in
  OUTPUT c3_stack_pair_timeseries.csv and c3_validation_summary.json

PHASE 6: Combined CAM+C3 meeting package
  DO make_cam_c3_professor_plots.py
  OUTPUT CAM main plot, C3 plot, combined comparison, talking points
```

3.2 Current Script Roles

Script	Role	Important output
run_cam_gsw_driven.py	Run CAM model with fixed inputs and gsw-driven / phase-aware assumptions.	outputs_cam_gsw_fixed_inputs/*.csv
green_roof_cam_gsw.py	Core model functions and v7 slab node export.	T_s_top, T_s_mid, T_s_in columns.
cam_stack_aware_validation.py	Compare measured sensors against physical stack model nodes.	cam_stack_validation_summary.json and cam_stack_pair_timeseries.csv.
make_cam_professor_plots.py	Create CAM-only professor-ready plots.	01_main_T2A_vs_Ts_in_validation.png and talking points.
run_c3_quick_validation.py	Create missing C3 validation output by scanning candidate sensors and model nodes.	outputs_c3_stack_validation/ c3_stack_pair_timeseries.csv.
make_cam_c3_professor_plots.py	Combine CAM and C3 validation plots for professor meeting.	CAM+C3 summary table and comparison plot.

Chapter 4: Current CAM Validation Finding

4.1 Main Finding

The latest stack-aware validation confirms that T_s_top is now exported and available, so the v7 patch worked. However, the physical stack comparison remains poor for several expected pairs. This means the remaining problem is not just missing output columns; it is a sensor/path mapping and heat-path representation issue.

Physical pair	Latest result	Interpretation
T1Ka vs T_in_used	RMSE approx 0.009 degC, corr approx 0.99997	Boundary input is correctly aligned.
T1Ke vs T_g_top	Model peak approx 56.37 degC vs measured peak approx 35.05 degC; corr approx -0.09	Top substrate model is too hot / wrong phase.
T2A vs T_g_bot	Bias-corrected RMSE approx 0.89 degC but corr approx 0.11	Amplitude close after bias correction, but shape/timing poor.
T1Ta vs T_s_top	Measured amp approx 16.58 degC; model amp approx 2.11 degC; corr approx -0.07	Slab-top model too damped or T1Ta not clean slab-top path.
T1Tb vs T_s_in	Measured amp approx 10.08 degC; model amp approx 1.99 degC; corr approx 0.35	T1Tb is too dynamic for current 1D T_s_in.

4.2 Why T2A Is Used Temporarily

Comparison	Bias-corrected RMSE	Correlation	Amplitude error	Decision
T2A vs T_s_in	approx 0.57 degC	approx 0.57	approx -0.25 degC	Best practical damped-path validation among predicted green-roof nodes.
T2A vs T_g_bot	approx 0.89 degC	approx 0.11	approx -0.08 degC	Amplitude close but timing/shape poor.
T2A vs T_in_used	approx 0.84 degC	approx 0.78	approx +1.85 degC	Not used as final target because T_in_used is input boundary, not predicted green-roof node.

Temporary conclusion
T2A should be presented as a temporary empirical damped-path validation target, not as confirmed physical T_s_in or confirmed lower substrate until plug-off test verifies the channel mapping.

4.3 Why T1Tb Is Diagnostic, Not Main Target

T1Tb was originally attractive because older documentation labels it as atap indoor. However, current data show that measured T1Tb has a much larger amplitude than model T_s_in. Bias correction cannot solve amplitude/path mismatch.

Metric	T1Tb measured	Model T_s_in	Meaning
Robust amplitude 95-05%	approx 10.08 degC	approx 1.99 degC	Model is far too flat.
Peak temperature	approx 38.17 degC	approx 30.53 degC	Peak model too low by about 7.64 degC.
Correlation	-	approx 0.35	Shape similarity is weak.
Bias correction effect	Small improvement only	Still poor	Problem is not only vertical offset.

Chapter 5: C3 Validation Plan

A C3 validation script was added because the combined CAM+C3 plotter needs a C3 validation output. The quick C3 validation is intended for professor-progress reporting, not final sensor proof.

5.1 Quick C3 Validation Logic

```
run_c3_quick_validation.py
  LOAD NI files
  LOAD C3 model output folder
  FILTER selected C3 time window
  SCAN candidate C3 sensors against available model nodes
  RANK candidates using:
      bias-corrected RMSE
      absolute correlation
      amplitude error
  CHOOSE target sensor for requested model node, default T_s_in
  OUTPUT c3_stack_pair_timeseries.csv, c3_validation_summary.json, matrix heatmaps
```

5.2 C3 Default Assumptions

Item	Default	Why
Default C3 window	2026-04-09 11:05 to 2026-04-10 14:08	Matches previously discussed C3 working window. Can be overridden.
Default target node	T_s_in	Main inside/damped thermal output for model comparison.
Candidate sensors	T3Ka, T3Kb, T3Kc, T3Kd, T3Ke, T2Ka, T2Kc, T2Kd, 2Ke, T1A	Conservative scan because C3 physical mapping is not yet fully locked.
If physical sensor is known	--target-sensor <channel>	Forces known target and avoids purely statistical selection.

Chapter 6: Formula & Numerical Model Reference

This chapter summarizes the formula chain used by the finite-volume simulation. Detailed equation numbering should be matched to the reference paper in the thesis methodology chapter.

6.1 Chronological Execution Order

Step	Expression / process	Role
F0	dz_sub = H_g/(N_g-1), dz_slab = H_slab/(N_s-1)	Finite volume grid spacing.
F1	Convert weather: T_a_C to K, rain to mass flux, G_sol in W/m2	Boundary forcing.
F2	P_sat(T), P_a, VPD, T_sky, psychrometric constant gamma	Thermodynamic helper variables.
F3	Aerodynamic resistance r_a from canopy height and wind speed	Controls sensible and latent exchange.
F4	Stomatal resistance r_stoma = (r_min/LAI) f_light f_temp f_VPD f_moisture	Plant transpiration control.
F5	Foliage energy balance: radiation + convection + latent exchange	Solves foliage temperature T_f.
F6	Substrate heat equation with moisture-dependent rho_cp and lambda	Solves T_g profile.
F7	Richards/moisture equation with rainfall, evapotranspiration, and drainage	Solves theta profile.
F8	Slab conduction with top continuity and bottom indoor convection	Solves T_s_top, T_s_mid, T_s_in.
F9	q_s_in = h_in (T_s_in - T_in)	Indoor heat flux output.
F10	Q_gain = time integral of q_s_in	Net heat gain over simulation period.

6.2 CAM vs C3 Physiological Difference in Model

Aspect	CAM	C3
Stomatal behavior	Daytime stomatal resistance increased; nighttime resistance may be lower.	Stomata generally more open in daytime if light/water conditions allow.
Evapotranspiration expectation	Lower daytime transpiration and weaker daytime cooling.	Higher daytime transpiration and stronger evaporative cooling.
Data caveat	CAM Licor data are discontinuous and sometimes show similar day/night 95th percentile gsw; use carefully.	C3 Licor data more directly available for r_stoma estimate.
Validation caveat	Current mismatch is dominated by sensor/path issues, not necessarily CAM physiology.	C3 quick validation is now added but still needs final physical mapping.

Chapter 7: Corrections and Current Decisions

ID	Decision / correction	Current status
D1	Dynamic T_in from NI sensor instead of constant 25 degC.	Keep. T1Ka vs T_in_used confirms alignment.
D2	Do not assume any sensor is directly exposed.	Keep. Use "most solar-responsive" wording only if needed.
D3	v7 exports T_s_top and T_s_mid in addition to T_s_in.	Done. Stack validation now reads T_s_top.
D4	Do not tune H_g, H_slab, LAI, cover fraction, rho_g, theta_sat casually just to force graph fit.	Keep. Physical parameters should be measured/fixed.
D5	Use T2A as temporary CAM damped-path validation target.	Current best practical target.
D6	Treat T1Tb as diagnostic because it is too dynamic for T_s_in.	Current interpretation.
D7	Treat T1Ta vs T_s_top mismatch as possible path/sensor issue.	Needs physical check.
D8	C3 validation script added after realizing no C3 validation output existed.	Done. Run before combined plots.
D9	Report raw and bias-corrected metrics separately.	Keep. Do not hide bias correction.
D10	Plug-off sensor test is required before final mapping.	TODO.
D11	Avoid claiming T2A is definitively a physical slab node.	Keep. Phrase as temporary empirical damped-path target.
D12	Prepare professor plots as progress, not final model proof.	Use for meeting.

Chapter 8: Parameter and Variable Gap Analysis

Parameter / variable	Current value / status	Source / comment	Action
H_f CAM, H_f C3	Measured previously	Plant height affects aerodynamic resistance.	Keep unless updated measurement exists.
LAI CAM, LAI C3	Measured/ImageJ/calibrated candidate	Important sensitivity parameter.	Use measured value; do not force-fit without justification.
cover_fraction	Measured/ImageJ	Used for canopy coverage interpretation.	Keep fixed once measured.
rho_f / tau_f / alpha_f	Measured or literature-based	Leaf optical properties.	Keep documented; do not over-tune.
r_stoma_min C3	From Licor estimate	Use C3 Licor data where valid.	Keep as measured estimate.
r_stoma_min CAM	From CAM Licor ranges / cautious estimate	CAM Licor not continuous and several sessions show low gsw.	Use as physiology assumption, not main fitting knob.
H_g	10 cm noted in conversation	Substrate depth.	Physically measure/confirm.
H_slab	Fixed physical geometry	Slab thickness.	Measure/confirm, do not calibrate blindly.
rho_g	Should be fixed	Bulk density of soil + rice husk.	Lab/estimate, then fix.
theta_sat	Should be fixed	Porosity/saturation water content.	Lab/estimate, then fix.
soil moisture	Short-window data available	S1 likely shallow and S2 deeper based on behavior, but verify.	Use for constraint/diagnostic; not full continuous forcing yet.
sensor mapping	Not final	T1Ta/T1Tb/T2A behavior does not fully match assumed physical stack.	Plug-off test.

Chapter 9: Professor Meeting Package

9.1 Plots to Show

Order	Plot	Message
1	01_CAM_main_T2A_vs_Ts_in.png	CAM can reproduce a damped response when T2A is used as temporary damped-path target.
2	02_CAM_diagnostic_T1Tb_vs_Ts_in.png	T1Tb is too dynamic for the current 1D T_s_in node, so it is not forced as final target.
3	04_CAM_C3_summary_table.png or stack summary table	Show quantitative progress and honest diagnostic results.
4	02_C3_validation.png after run_c3_quick_validation.py	Show C3 comparison once quick validation is generated.

9.2 Suggested Meeting Wording

Suggested wording in Indonesian

Pak, saya awalnya memvalidasi CAM terhadap sensor atap indoor T1Tb, tetapi hasil model terlihat terlalu flat. Setelah saya lakukan stack-aware validation, T1Tb ternyata memiliki amplitudo jauh lebih besar daripada node T_s_in model 1D, sehingga kemungkinan ada perbedaan path panas atau mapping sensor yang belum terkonfirmasi. Untuk update sementara, saya gunakan T2A sebagai target validasi damped thermal response karena bentuk dan amplitudonya lebih dekat dengan node model. Setelah ini saya akan lakukan plug-off test sensor untuk mengunci mapping fisik sebelum finalisasi validasi.

9.3 What Not to Claim

Avoid saying	Safer wording
The model is fully validated.	The model shows promising damped-path agreement for T2A, but sensor mapping still requires verification.
T2A is definitely T_s_in.	T2A is used as a temporary empirical damped-path target.
T1Tb is wrong.	T1Tb does not behave like a clean 1D T_s_in node in the current data.
The problem is CAM stomata.	The current main issue appears to be sensor/path mapping and thermal-path representation.

Chapter 10: Quick Reference

10.1 Commands

```
# 1) Run CAM v7 model
rmdir /s /q __pycache__
python run_cam_gsw_driven.py

# 2) Run CAM stack-aware validation
python cam_stack_aware_validation.py

# 3) Make CAM-only professor plots
python make_cam_professor_plots.py

# 4) Run C3 quick validation
python run_c3_quick_validation.py

# Optional C3 custom window
python run_c3_quick_validation.py --start "2026-04-09 11:05:00" --end "2026-04-10 14:08:00"

# Optional C3 known target sensor
python run_c3_quick_validation.py --target-sensor T3Ka

# 5) Make combined CAM+C3 professor plots
python make_cam_c3_professor_plots.py
```

10.2 Output Files

Folder	Important files
outputs_cam_gsw_fixed_inputs	CAM model prediction CSV with T_s_top, T_s_mid, T_s_in.
outputs_cam_stack_validation	cam_stack_validation_summary.json, cam_stack_pair_timeseries.csv, stack-aware plots.
outputs_prof_meeting_cam	CAM-only professor plots and talking points.
outputs_c3_stack_validation	c3_stack_pair_timeseries.csv, c3_validation_summary.json, target matrix.
outputs_prof_meeting_cam_c3	Combined CAM+C3 plots, summary table, talking points.

10.3 Decision Tree

```
IF professor asks: Why not T1Tb?
  Answer: T1Tb amplitude is too high compared with model T_s_in; likely path/sensor issue.

IF professor asks: Why T2A?
  Answer: It is the best temporary damped-path target among predicted green-roof nodes after bias
correction.

IF professor asks: Is the model final?
  Answer: Not yet. Current update is path-aware validation and sensor diagnostic. Next is plug-off
test.

IF professor asks: What is next?
  Answer: Confirm sensor mapping physically, then rerun stack-aware validation and finalize target
pairs.

IF professor asks: Why C3 is not included yet?
  Answer: C3 quick validation script is now added; it scans target sensors and prepares comparison
plots.
```

End of Current Working Documentation
Green Roof Thermal Simulation - CAM vs C3
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